# Distributivity and the Commensurability of Value-Definite Realism and Empirical Adequacy

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**Abstract**

When does an empirically adequate theory admit a value-definite completion — an assignment of simultaneous definite values to all observables? The answer depends on the algebraic structure of the observations. For a directed system of Boolean algebras carrying compatible charges, the Stone construction produces an unconditional measure on the dual space, and every observable admits a simultaneous definite value via ultrafilters. The passage to a probability measure on an underlying state space turns on $\sigma$-additivity of the charges, together with a Loomis–Sikorski realization condition on the choice of state space.

For orthomodular lattices — the algebraic framework for incompatible observations — the situation is structurally different. The McDonald–Bimbó dual space carries principal filters as distinguished structure, and the Kochen–Specker theorem blocks global value-definiteness. Probabilistic realism splits into two grades: Gleason-type results supply a $\sigma$-additive measure on the *lattice* ($\mathrm{PR}_{\mathrm{lat}}$), but the passage to a measure on the *dual space* ($\mathrm{PR}_{\mathrm{dual}}$) fails — for the central case $L(H)$, no normal state extends to a charge on the dual, so even probabilistic realism in the dual-space sense is obstructed, and value-definite realism fails outright.

Distributivity guarantees value-definiteness; for the physically central case $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. The state space admits a natural filtration $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\mathrm{df}}$, whose three transitions — pasting, regularity, sharpness — are of different mathematical character. Distributivity collapses the filtration; non-distributivity can (but need not) open it.

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## 1 Introduction

A central question in the philosophy of science asks when an empirically successful theory can be completed to a realist one — one that assigns definite properties to an underlying reality, not merely adequate predictions to observable phenomena. Van Fraassen [37] argues that science need only aim at *empirical adequacy*: agreement with all observable phenomena. The scientific realist demands more — an underlying state space whose properties generate the observations.

This debate is usually conducted in philosophical terms. We give it a mathematical formulation.

Three nested positions, each a strengthening of the previous:

(i) **Empirical adequacy (EA).** A measure on the dual space of the observation algebra that agrees with all observed charges. No commitment to an underlying state space.

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(ii) **Probabilistic realism (PR).** A probability measure on an underlying state space $(\Omega, \sigma(\mathcal{C}))$ that generates the observations. Requires descent from the dual-space measure.

(iii) **Value-definite realism (VDR).** Every observable simultaneously has a definite value: a global dispersion-free state on the full observation algebra. On a Boolean algebra (and on $L(H)$) this coincides with a 2-valued homomorphism; on a general OML the two come apart (Remark below), and it is the dispersion-free state that is the operative notion throughout.

Both EA and PR as stated presuppose a genuine measure on the dual space: EA *is* such a measure, and PR descends from it. In the Boolean case this presupposition is free (the Stone construction supplies the dual-space measure unconditionally). In the OML case it is not, and PR accordingly refines into two grades — $\mathrm{PR}_{\mathrm{lat}}$, a measure on the lattice, and $\mathrm{PR}_{\mathrm{dual}}$, a measure on the dual space concentrating on the physical points — which coincide for Boolean algebras and separate for $L(H)$ (Theorem 3.1). Where the distinction matters we write the grade explicitly; an unqualified "PR" refers to the Boolean case, where the two agree.
These are nested: VDR implies PR implies EA. Which implications reverse depends on the observation algebra:

*Compatible observations.* When all observations can be performed simultaneously, the observation algebra is a Boolean algebra — a complemented distributive lattice. Every Boolean algebra admits 2-valued homomorphisms (ultrafilters exist by Zorn's lemma). In this setting, all three positions are available and mutually commensurable. The passage from EA to PR requires $\sigma$-additivity of the charges (and a realization condition on $\Omega$, developed below); the further passage to VDR is free.

*Incompatible observations.* When some observations cannot be performed simultaneously — as in quantum mechanics, contextual experimental design, or distributed systems with interfering sensors[1] — the observation algebra is an orthomodular lattice (OML), a complemented but non-distributive lattice. The Kochen–Specker theorem [26] shows that for the prototypical case $L(H)$ with $\dim H \geq 3$, no 2-valued homomorphism exists: VDR is blocked. Probabilistic realism on the *lattice* remains available via Gleason's theorem [17], but its *dual-space* form fails for normal states (no extension to a charge on the McDonald–Bimbó dual space $S_0(L(H))$, Definition 2.9). Empirical adequacy itself is hedged in the OML case: it is the orthogonally additive set function on the $\perp$-stable clopens, which always exists, rather than a full measure on the dual space — promotion to such a measure is the extension step that fails for $L(H)$.

Distributivity guarantees value-definiteness; for $L(H)$ with $\dim H \geq 3$, its absence blocks it. Every ingredient is known. The mathematical ones are Stone duality, the Kochen–Specker theorem, Gleason's theorem, the Bub–Clifton uniqueness result, and the recent McDonald–Bimbó duality for orthomodular lattices [29]. The philosophical architecture is older still: the distinction between a probabilistic state and a value-definite one is van Fraassen's own dynamical-state/value-state distinction [38], elaborated throughout the modal-interpretation literature [3, 39, 6] — our EA, PR, and VDR *formalise, rather than originate*, that hierarchy. That value-definiteness is separable from realism and rests on the Boolean structure of jointly performable observations is Stairs [36]; that the realism question turns on distributivity is the Putnam [33]–Dummett dispute; that the Boolean/non-Boolean event structure is primitive is Bub–Pitowsky [5]. In the von Neumann algebra setting, the separation between a state qua

[1] The non-quantum examples require the incompatibility to be *operationally imposed*: the contexts must be genuinely non-co-realizable, with no joint probability space carrying all observations at once. This is the content of contextuality as failure of a global section (Abramsky–Brandenburger [1]; Fine's theorem [16] for the joint-distribution criterion). Incompatibility cannot arise *intrinsically* from the dynamics of a single classical system: any finite family of classical observables on a common probability space jointly distributes (Kolmogorov), hence sits inside a Boolean algebra. Distributed-sensor contextuality is therefore a statement about non-co-realizable measurement settings, not about the dynamics of one underlying process.

measure on the lattice and the Kochen–Specker absence of global points is the content of the topos program — Isham–Butterfield's reading of Kochen–Specker as the non-existence of a global section [24] and Döring's measures on the spectral presheaf [11] — and the question whether such dual-space measures amount to realism has been pressed directly by Eva [15].

What we add is neither a new philosophical thesis nor a new theorem but a *bridge*: we carry this separation across the McDonald–Bimbó duality — a quantum-logic result with no prior contact with the philosophy-of-science literature — read it against the van Fraassen empirical-adequacy/realism debate, and organise EA, $\mathrm{PR}_{\mathrm{lat}}$, $\mathrm{PR}_{\mathrm{dual}}$, and VDR onto a single axis indexed by distributivity. The dual-space measures of the topos program live on the spectral presheaf [11, 7] — a diagram over the category of commutative contexts; ours live on the McDonald–Bimbó dual space $S_0(A)$, a single topological space carrying the principal filters $P(A)$ as distinguished structure. It is in those terms — a measure on $S_0(A)$ concentrating on $P(A)$ — that the $\sigma$-additive passage to the dual space, and its failure for $L(H)$, has not previously been isolated. We begin with the Boolean case.[2]

Let $(\iota, \leq)$ be a directed set indexing a family of Boolean algebras $\{B_i\}_{i\in\iota}$ with injective homomorphisms $\varphi_{ij} : B_i \to B_j$ for $i \leq j$. Write $\mathcal{C} = \bigcup_i B_i$ for the direct limit — the *cylinder algebra.*

A *compatible family of normalised charges* is a collection $\{\ell_i\}$ with each $\ell_i : B_i \to [0, 1]$ finitely additive, $\ell_i(1) = 1$, and

$$\ell_j \circ \varphi_{ij} = \ell_i \qquad (i \leq j).$$

The Stone space $\mathrm{St}(\mathcal{C})$ is the space of ultrafilters on $\mathcal{C}$, compact and totally disconnected. Each $\omega \in \Omega$ determines a principal ultrafilter

$$\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}.$$

Any compatible family of normalised charges extends to a $\sigma$-additive Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ — unconditionally, with no assumption of $\sigma$-additivity. Each charge extends to a regular Baire measure on the compact, totally disconnected $\mathrm{St}(B_i)$: the clopens are compact, so any countable clopen cover of a clopen set has a finite subcover, making a finitely additive charge automatically countably additive on the clopen algebra (a premeasure), which Carathéodory extends [34, 19]; compatibility then assembles these into $\hat{\mu}$ on the inverse limit $\mathrm{St}(\mathcal{C})$ [9, Theorem 1]. This is the *empirically adequate* object: it agrees with all charges on cylinder events.

The measure $\hat{\mu}$ lives on $\mathrm{St}(\mathcal{C})$, not on $\Omega$. Descent to a probability measure $P$ on $(\Omega, \sigma(\mathcal{C}))$ requires

$$\hat{\mu}\big(\mathrm{pure}(\Omega)\big) = 1,$$

i.e., that $\hat{\mu}$ concentrates on the principal ultrafilters. Such concentration implies each $\ell_i$ is $\sigma$-additive; the converse holds when $\Omega$ realizes every countably-complete ultrafilter of $\mathcal{C}$, a Loomis–Sikorski condition that is automatic for countably generated systems but not in general.

Every Boolean algebra admits lattice ultrafilters — maximal proper filters — by Zorn's lemma. On a Boolean algebra, maximality forces a dichotomy: for every $a$, either $a \in F$ or $a^c \in F$. The proof extends $F$ to include $a^c$ whenever $a \notin F$; the extension remains proper because $a^c \wedge f \neq \perp$ for all $f \in F$, which follows from distributivity. This dichotomy makes every lattice ultrafilter a 2-valued homomorphism, assigning each $a \in \mathcal{C}$ a definite value in $\{0, 1\}$. On a non-distributive OML, the extension argument fails, and maximal proper filters need not be 2-valued.

Moreover, any subset $S \subseteq \mathrm{St}(\mathcal{C})$ that projects surjectively onto each $\mathrm{St}(B_i)$ can serve as a realisation space: set $\Omega = S$ with evaluation maps given by the restriction maps. The algebra imposes no constraint on the choice of $\Omega$.

[2] The Boolean construction is self-contained here; a fuller treatment of the directed-system Stone extension and the realization condition is developed in [28].

**Proposition 1.1** (Boolean commensurability)**.** *For a directed system of Boolean algebras with compatible charges:*

*(a) EA is unconditional:* $\hat{\mu}$ *on* $\mathrm{St}(\mathcal{C})$ *always exists.*

*(b) PR requires each* $\ell_i$ *to be* $\sigma$*-additive, and is equivalent to it under the Loomis–Sikorski realization condition on* $\Omega$ *(automatic for countably generated systems).*

*(c) VDR holds unconditionally:* $\mathrm{St}(\mathcal{C})$ *consists entirely of 2-valued homomorphisms.*

*The obstruction at the EA–PR transition is* $\sigma$*-additivity of the charges, together with the Loomis–Sikorski realization condition on* $\Omega$ *(automatic for countably generated systems); both are conditions on the charges and the realization, not on the algebra.*

*Proof.* Part (a) is the Stone construction above [34, 9]. Part (b): descent to $\Omega$ requires $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$, which forces each $\ell_i$ $\sigma$-additive, and conversely under the realization condition (discussion above). Part (c): ultrafilters on a Boolean algebra are exactly the 2-valued homomorphisms; their existence follows from Zorn's lemma applied to the filter generated by any non-zero element. □

# 2 Orthomodular observation algebras

Proposition 1.1 depends on distributivity at exactly one point: the existence of 2-valued homomorphisms. When observations are incompatible, this fails. The non-distributive structure need not be postulated: Gunji et al. [18] show that gluing Boolean algebras along shared boundaries via categorical pushout produces an orthomodular lattice, generically non-distributive. The OML is *forced* by the context structure.

**Definition 2.1.** An *ortholattice* is a bounded lattice $A = (A, \wedge, \vee, {}^\perp, 0, 1)$ with an orthocomplementation ${}^\perp : A \to A$ satisfying $a \wedge a^\perp = 0$, $\quad a \vee a^\perp = 1$, $\quad a^{\perp\perp} = a$, and $\quad a \leq b \Rightarrow b^\perp \leq a^\perp$.

**Definition 2.2.** An *orthomodular lattice* (OML) is an ortholattice satisfying the *orthomodular law*:

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b).$$

Every Boolean algebra is an OML (the orthomodular law follows from distributivity), but the converse fails. An OML is Boolean if and only if it is distributive [29, Proposition 2.3].

**Example 2.3.** The lattice $L(H)$ of closed linear subspaces of a separable Hilbert space $H$ with $\dim H \geq 3$, ordered by inclusion with $U^\perp = \{x \in H : \langle x, y\rangle = 0 \ \forall y \in U\}$, is an orthomodular lattice that is not distributive [25, Chapter 1].

**Example 2.4** (Incompatible measurements)**.** Three binary measurements $A$, $B$, $C$, where $(A, B)$ and $(A, C)$ can be performed jointly but $(B, C)$ cannot. The jointly measurable pairs generate Boolean subalgebras; the full observation algebra is orthomodular, not Boolean.

Two elements $a, b$ in an OML *commute* if they generate a Boolean subalgebra. A maximal set of pairwise commuting elements forms a *Boolean context* — a maximal Boolean subalgebra of $A$.

**Definition 2.5.** A *state* on an OML $A$ is a function $s : A \to [0, 1]$ with $s(1) = 1$ and

$$a \perp b \quad \Longrightarrow \quad s(a \vee b) = s(a) + s(b),$$

where $a \perp b$ means $a \leq b^\perp$.

A state is additive on *orthogonal* pairs only — not on arbitrary joins. This is the OML analogue of a finitely additive charge on a Boolean algebra.

**Definition 2.6.** A state $s$ is *dispersion-free* if $s(a) \in \{0,1\}$ for all $a \in A$.

A dispersion-free state assigns a definite truth value to every proposition in $A$ — the statistical/dispersion-free distinction in the standard quantum-logic sense [2]. Every 2-valued homomorphism is dispersion-free; the converse holds on Boolean algebras (where dispersion-free states are exactly the ultrafilters) and on $L(H)$, but *fails* on a general OML: a dispersion-free state need not be multiplicative across incompatible propositions. The minimal witness is $\mathrm{MO}_2$ (the horizontal sum of two Boolean $2^2$ blocks): assigning 1 to one atom of each block gives a dispersion-free state $s$, yet for the gap pair $a, b$ with $a \wedge b = 0$ and $s(a) = s(b) = 1$ one has $s(a \wedge b) = 0 \neq 1 = s(a) \wedge s(b)$, so $s$ is not a homomorphism. Throughout, the operative notion of value-definiteness is the dispersion-free state (equivalently, non-emptiness of $\mathcal{S}_{\mathrm{df}}(A)$), not the strictly stronger 2-valued homomorphism.

**Theorem 2.7** (Kochen–Specker [26])**.** *If* $\dim H \geq 3$*, there is no dispersion-free state on* $L(H)$*.*

For $\dim H = 2$, dispersion-free states exist; the obstruction is specific to $\dim H \geq 3$.

**Theorem 2.8** (Gleason [17])**.** *If $H$ is a separable Hilbert space with* $\dim H \geq 3$*, every $\sigma$-additive state on $L(H)$ is of the form* $s(P) = \mathrm{tr}(\rho P)$ *for a unique density operator $\rho$ on $H$.*

The analogue of Stone duality for OMLs is the McDonald–Bimbó duality [29].

Let $A$ be an OML. A *filter* on $A$ is a non-empty subset $x \subseteq A$ that is upward closed ($a \in x$ and $a \leq b$ implies $b \in x$) and closed under finite meets ($a, b \in x$ implies $a \wedge b \in x$). A filter is *proper* if $0 \notin x$; the unique *improper* filter is $\omega = A$.

For each $a \in A$, the *principal filter* generated by $a$ is $\uparrow a = \{b \in A : a \leq b\}$. Write $F(A)$ for the collection of all filters on $A$, and $P(A) \subseteq F(A)$ for the subcollection of principal filters.

**Definition 2.9** (McDonald–Bimbó [29])**.** Let $A$ be an OML. The *dual space* of $A$ is the ordered relational topological space

$$S_0(A) = \big(F(A),\ \subseteq,\ \perp_A,\ P(A),\ \mathcal{T}\big),$$

where:

(i) $F(A)$ is the set of all filters on $A$;

(ii) $\subseteq$ is set inclusion;

(iii) $x \perp_A y$ iff there exists $a \in A$ with $a \in x$ and $a^\perp \in y$;

(iv) $P(A)$ is the collection of principal filters;

(v) $\mathcal{T}$ is the topology generated by the subbasis $\{h(a) : a \in A\} \cup \{F(A) \setminus h(a) : a \in A\}$, where $h(a) = \{x \in F(A) : a \in x\}$.

**Theorem 2.10** (McDonald–Bimbó [29])**.** *$S_0(A)$ is an orthomodular space (compact, totally disconnected). The category of OMLs and homomorphisms is dually equivalent to the category of orthomodular spaces and continuous weak p-morphisms. Moreover,* $A \cong \mathrm{CO}(S_0(A))^\dagger$*, the OML of clopen $\perp$-stable subsets of $S_0(A)$.*

A subset $U \subseteq S_0(A)$ is $\perp$-*stable* if $U = U^{\perp\perp}$, where $U^\perp = \{x : x \perp_A y \text{ for all } y \in U\}$ — i.e. $U$ is closed under the double dual-orthocomplement. We write $\mathrm{CO}(S_0(A))^\dagger$ for the sub-OML of clopen $\perp$-stable subsets; its meet is intersection and its complement the dual orthocomplement, but its join is *not* set union.

The critical differences from Stone duality:

| Feature | Boolean (Stone) | OML (McDonald–Bimbó) |
|---|---|---|
| Dual points | Ultrafilters | All filters |
| Distinguished subset | None | $P(A)$ (principal filters) |
| Clopens | Boolean algebra | OML |
| Orthogonality | Trivial | Non-trivial ($\perp_A$) |
| 2-valued homomorphisms | Plentiful (Zorn) | May not exist (KS) |

In Stone duality, the principal filters (i.e., the principal ultrafilters pure($\omega$) for $\omega \in \Omega$) are *invisible*: they play no distinguished role in the dual space. Which ultrafilters are principal depends on the choice of $\Omega$.

In the McDonald–Bimbó duality, $P(A)$ is *part of the structure* of the dual space: the orthomodular frame conditions (Definition 2.9) explicitly reference $P(A)$. This is the algebraic formalisation: for non-distributive algebras, the distinction between realised and unrealised observation-patterns is constrained by the algebra. Distributivity is the condition under which this constraint vanishes.

## 2.1 Descent to the dual space and its obstruction

$S_0(A)$ is compact, so the topological machinery applies. But the algebraic input is weaker. A state $s$ on $A$ defines

$$\mu(h(a)) = s(a)$$

on $\mathrm{CO}(S_0(A))^\dagger$.

In the Boolean case, $\mathrm{Clop}(\mathrm{St}(\mathcal{C}))$ is a Boolean algebra of sets, and $s$ is finitely additive on it. The Carathéodory–Choksi extension machinery applies: compactness of $\mathrm{St}(\mathcal{C})$ gives $\sigma$-subadditivity, and the extension to a Baire measure is automatic.

In the OML case, $\mathrm{CO}(S_0(A))^\dagger$ is an OML of sets — closed under $\cap$ and $^\perp$ but *not* under arbitrary $\cup$. The state $s$ is additive on orthogonal pairs only. Carathéodory does not apply: there is no Boolean algebra of sets on which $s$ is finitely additive.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem (Theorem 2.8) supplies a $\sigma$-additive measure *on the lattice* $L(H)$ for every $\sigma$-additive state: every such state arises from a density operator via the Born rule. This is a measure on $A$ itself, and is *not* the same as a measure on the dual space $S_0(A)$ — a distinction the Boolean case obscures and the OML case forces. Indeed the passage to the dual space *fails* for $L(H)$: no normal (Gleason) state extends to a charge on the clopens of $S_0(L(H))$. This is a finitary non-extendability of the Kochen–Specker [26] and Pitowsky correlation-polytope [30] type, whose mechanism we sketch rather than prove in full.[3] For general OMLs the extension problem is open.[4]

The conceptual separation between a measure on the lattice and the non-existence of a measure concentrated on points is, in the von Neumann algebra setting, already present in

[3]Within a single 2-dimensional subspace $e$, the lines $p$ and their orthocomplements-within-$e$ furnish, for every $k$, $2k$ pairwise meet-zero elements whose $\perp$-stable clopen images are pairwise disjoint, while orthoadditivity gives $s(p) + s(e \ominus p) = s(e)$. A charge would then force $k\,s(e) \leq 1$ for all $k$, so $s(e) = 0$ for every finite-dimensional $e$ — excluding every normal state. This is the 2-dimensional case of the Pitowsky correlation-polytope condition [30]; $\sigma$-completeness of the lattice is irrelevant, the obstruction being finitary. Only *singular* states ($s(e) = 0$ on all finite-dimensional $e$) escape, and whether one extends is open.

[4]The extension splits into two axes that the Boolean case fuses. The *extension axis* — passing from the orthogonally additive state on the OML of $\perp$-stable clopens to a charge on the full Boolean algebra of clopens of $S_0(A)$ — is the classical Boolean extension problem [23]: since the representation map preserves meets, meet-zero elements map to disjoint sets, so the condition is $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families, a Pitowsky polytope condition [30] strictly stronger than orthoadditivity when the OML is non-distributive. It is not governed by $\sigma$-additivity, and in the finite case is a decidable linear program. The *descent axis* — whether the resulting measure concentrates on the physical points — is what $\sigma$-additivity governs. For $L(H)$, Gleason settles the lattice measure, but the extension axis itself fails (preceding footnote): Gleason does *not* deliver the dual-space measure.

Döring's analysis of the spectral presheaf [11]: states correspond to measures on the clopen subobjects of the presheaf $\Sigma$ ($\mathrm{PR}_{\mathrm{lat}}$ in our terms), while the absence of global sections of $\Sigma$ is the Kochen–Specker obstruction ($\mathrm{PR}_{\mathrm{dual}}$ failing). Our contribution is not this separation as such but its transposition: we phrase it through the McDonald–Bimbó duality for orthomodular lattices rather than the topos-theoretic presheaf, identify $\sigma$-additivity as the precise marker of the descent axis (Döring's measures are finitely additive in general), and read the whole off against the van Fraassen empirical adequacy/realism debate, a framing absent from the topos literature.

Even when probabilistic descent succeeds, value-definite descent does not. The Bub–Clifton theorem [4] addresses what *can* be salvaged:

**Theorem 2.11** (Bub–Clifton [4]; Halvorson–Clifton [20])**.** *Given a state $\rho$ on $L(H)$ and a preferred observable $R$, there is a unique maximal sublattice $D(\rho, R) \subseteq L(H)$ such that:*

*(i) $D(\rho, R)$ is a Boolean algebra;*

*(ii) $D(\rho, R)$ contains the spectral projections of $R$;*

*(iii) The restriction of $\rho$ to $D(\rho, R)$ is a mixture of dispersion-free states.*

In the present vocabulary, condition (iii) says that $D(\rho, R)$ is the maximal Boolean subalgebra on which VDR is available — restating Halvorson and Clifton [20] in the EA/PR/VDR framework. It depends on $R$: different preferred observables yield different maximal Boolean subalgebras.

## 3 Distributivity as the commensurability condition

The status of the three positions depends on the algebra. In the OML case probabilistic realism splits into two grades that the Boolean case fuses: $\mathrm{PR}_{\mathrm{lat}}$, a $\sigma$-additive measure on the *lattice* $A$ — states as measures on the projection lattice in the sense of the algebraic quantum-logic tradition [35] (Gleason, for $L(H)$) — and $\mathrm{PR}_{\mathrm{dual}}$, a measure on the *dual* $S_0(A)$ concentrating on the physical points. We take $\mathrm{PR}_{\mathrm{dual}}$ to be the *stronger* grade — a dual-space measure restricts, on the $\perp$-stable clopens, to a finitely orthoadditive set function recovering $s$; promoting that to a $\sigma$-additive lattice measure ($\mathrm{PR}_{\mathrm{lat}}$) is free in the Boolean case but only conditional in general (see $(*)$). The relationship the two grades certainly exhibit is a *witnessed separation*, not a proven nesting: for $L(H)$, $\mathrm{PR}_{\mathrm{lat}}$ holds while $\mathrm{PR}_{\mathrm{dual}}$ fails. The positions, ordered with the stronger to the right (a node marked † is not realised for $L(H)$):

$$\underbrace{\mathrm{EA}}_{\perp\text{-stable set fn.}} \quad \Longleftarrow \quad \underbrace{\mathrm{PR}_{\mathrm{lat}}}_{\text{Gleason } (L(H))} \quad \overset{(*)}{\Longleftarrow} \quad \underbrace{\mathrm{PR}^{\dagger}_{\mathrm{dual}}}_{\text{fails for } L(H)} \quad \Longleftarrow \quad \underbrace{\mathrm{VDR}^{\dagger}}_{\text{KS-blocked (OML)}}$$

The implications run weak-from-strong: VDR (a global dispersion-free state) implies $\mathrm{PR}_{\mathrm{dual}}$ (a dual-space measure); $\mathrm{PR}_{\mathrm{dual}}$ implies $\mathrm{PR}_{\mathrm{lat}}$ only conditionally (the restriction of $\hat{\mu}$ to the $\perp$-stable clopens is finitely orthoadditive, but its $\sigma$-additivity on the lattice needs $\hat{\mu}$ to be null on countable-join defects — free in the Boolean case, conditional in general; see $(*)$); and $\mathrm{PR}_{\mathrm{lat}}$ implies EA. For $L(H)$ the two strongest positions are not realised: VDR by Kochen–Specker, and $\mathrm{PR}_{\mathrm{dual}}$ by the Kochen–Specker / Pitowsky non-extendability of §2.1 — the *upward* extension construction $\mathrm{PR}_{\mathrm{lat}} \to \mathrm{PR}_{\mathrm{dual}}$ (build a dual-space measure from the lattice measure) is obstructed, governed by distributivity rather than $\sigma$-additivity. Gleason delivers $\mathrm{PR}_{\mathrm{lat}}$ but Gleason does not climb to $\mathrm{PR}_{\mathrm{dual}}$. The two grades coincide only in the Boolean case, where the extension is free and the ladder collapses to the single EA $\Leftarrow$ PR of the Boolean case. $(*)$ The arrow $\mathrm{PR}_{\mathrm{lat}} \Leftarrow \mathrm{PR}_{\mathrm{dual}}$ is free in the Boolean case but holds only conditionally in general (it requires a $\sigma$-continuity hypothesis on the dual measure that the finitary McDonald–Bimbó duality does not supply); we use it only in the direction stated. EA in the OML case is the $\perp$-stable clopen set function (the hedged reading below), not itself a dual-space measure.

**Theorem 3.1** (Commensurability)**.** *Let $A$ be a complemented lattice.*

*(a) If $A$ is distributive (Boolean), then EA, PR, and VDR are mutually commensurable:*

- *EA is unconditional (Stone construction).*
- *PR requires the charges to be $\sigma$-additive, and is equivalent to it under the Loomis–Sikorski realization condition on $\Omega$. Here $PR_{lat}$ and $PR_{dual}$ coincide: the $\perp$-stable extension step is free, so a lattice measure and a dual-space measure stand or fall together.*
- *VDR holds unconditionally (ultrafilters exist).*

*Moreover, the choice of realisation space $\Omega$ is unconstrained by the algebra.*

*(b) If $A$ is non-distributive (OML with $A \cong L(H)$, $\dim H \geq 3$), then:*

- *EA is unconditional* in the $\perp$-stable set-function sense.[5]
- *$PR_{lat}$ holds (Gleason, Theorem 2.8): a $\sigma$-additive measure on the lattice $L(H)$.*
- *$PR_{dual}$* fails *for every normal state: no charge on $S_0(L(H))$* extending *$s$ exists (Kochen–Specker / Pitowsky non-extendability, §2.1), so a fortiori none concentrating on $P(A)$. Equivalently, the extension step $PR_{lat} \to PR_{dual}$ is obstructed — the one transition the Boolean case leaves free.*
- *VDR fails globally (Kochen–Specker, Theorem 2.7).*
- *VDR holds locally within each Boolean context (Bub–Clifton, Theorem 2.11).*

*Proof.* Part (a) is Proposition 1.1. For part (b): $\mathrm{PR}_{\mathrm{lat}}$ follows from Gleason's theorem (Theorem 2.8), which supplies a $\sigma$-additive measure on the lattice $L(H)$ for any $\sigma$-additive state. $\mathrm{PR}_{\mathrm{dual}}$ fails for normal states by the Kochen–Specker / Pitowsky non-extendability of §2.1: no charge on the clopens of $S_0(L(H))$ *extending* $s$ exists, so no such Baire measure on $S_0(L(H))$ and a fortiori none concentrating on $P(A)$. EA holds in the $\perp$-stable set-function sense unconditionally — the assignment $\mu(h(a)) = s(a)$ on $\mathrm{CO}(S_0(A))^\dagger$ always exists; note this does *not* go via $\mathrm{PR}_{\mathrm{dual}}$ (which fails), since for $L(H)$ the genuine dual-space measure does not exist. VDR-failure is from Kochen–Specker; local VDR from Bub–Clifton. □

**Corollary 3.2.** *Distributivity is sufficient for VDR. For the physically central case $A = L(H)$ with $\dim H \geq 3$, non-distributivity blocks VDR.*

*Proof.* Sufficiency: Theorem 3.1(a). Blockage: Theorem 3.1(b). □

**Remark 3.3.** The dimensional restriction is necessary. For $\dim H = 2$, $L(\mathbb{C}^2)$ is non-distributive but admits dispersion-free states. More generally, horizontal sums of Boolean algebras and certain finite OMLs are non-distributive yet admit dispersion-free states [25, Chapter 4] — though, as $\mathrm{MO}_2$ shows, these dispersion-free states need not be 2-valued homomorphisms. The sharp characterisation for general OMLs is: VDR is available iff $A$ admits a dispersion-free state, which is a property of $A$, not of distributivity alone. Distributivity is sufficient (ultrafilters exist) but not necessary in full generality. For the physically and epistemologically central case $\dim H \geq 3$, it is the effective dividing line.

The containment $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\mathrm{df}}(A)$ is standard in quantum logic [32, 25]. What we add is the identification of three transitions of different mathematical character, and the observation that distributivity controls which are trivial.

[5] EA here is the orthogonally additive set function $\mu(h(a)) = s(a)$ on the $\perp$-stable clopens, which always exists. It is *not* a genuine Baire measure on $S_0(A)$: promoting it to one is the extension step, which for $L(H)$ *fails* for every normal state (§2.1) and for general OMLs is open. The unconditional claim is thus about the clopen set function, not the dual-space measure.

**Definition 3.4** (Coherence filtration)**.** Let $A$ be a complemented lattice with a family of charges. The *state space* of $A$ admits a descending filtration:

$$\mathcal{S}(A) \;\supseteq\; \mathcal{S}_\sigma(A) \;\supseteq\; \mathcal{S}_{\mathrm{df}}(A),$$

where:

- $\mathcal{S}(A)$ is the set of states on $A$ (contextual coherence);
- $\mathcal{S}_\sigma(A) \subseteq \mathcal{S}(A)$ is the subset of $\sigma$-additive states (probabilistic coherence);
- $\mathcal{S}_{\mathrm{df}}(A) \subseteq \mathcal{S}_\sigma(A)$ is the subset of dispersion-free states (value-definite coherence).

Beneath the filtration sits a local condition: *consistency* requires only that each maximal Boolean subalgebra of $A$ individually admits a compatible state.

Each inclusion can be strict. The three transitions have different mathematical character:

| Transition | Character | Key | Obstruction |
|---|---|---|---|
| Consistency $\to \mathcal{S}(A)$ | Pasting (topological) | Do local sections glue? | Contextuality |
| $\mathcal{S}(A) \to \mathcal{S}_\sigma(A)$ | Regularity (analytic) | Does the state have good limits? | $\sigma$-additivity |
| $\mathcal{S}_\sigma(A) \to \mathcal{S}_{\mathrm{df}}(A)$ | Sharpness (algebraic) | Can probabilities collapse to $\{0,1\}$? | Kochen–Specker |

Distributivity collapses the filtration. For a Boolean algebra:

- *Pasting* is trivial: every local assignment extends to a global state (distributivity guarantees compatibility).
- *Sharpness* is free: ultrafilters exist (Zorn's lemma), so $\mathcal{S}_{\mathrm{df}}(A) \neq \varnothing$ unconditionally.
- The only non-trivial transition is *regularity*: $\sigma$-additivity of the charges (with the realization condition on $\Omega$), as developed above.

For an OML with $\dim \geq 3$, every transition is non-trivial:

- *Pasting* is non-trivial: contextuality obstructions are possible.
- *Regularity* requires Gleason-type results. (This filtration grades *internal* states of $A$ — i.e. $\mathrm{PR}_{\mathrm{lat}}$, the lattice measure, which Gleason supplies for $L(H)$; the further passage to a measure on the dual space $S_0(A)$, namely $\mathrm{PR}_{\mathrm{dual}}$ — the extension step, free in the Boolean case but blocked by non-distributivity — is a separate question that for $L(H)$ *fails* for normal states and for general OMLs is open.)
- *Sharpness* is blocked: $\mathcal{S}_{\mathrm{df}}(A) = \varnothing$ (Kochen–Specker).

Distributivity is sufficient to collapse the filtration. For $L(H)$ with $\dim \geq 3$, non-distributivity opens it fully.

**Remark 3.5** (Relation to contextuality hierarchies)**.** Abramsky and Brandenburger [1] introduced a hierarchy of contextuality — probabilistic, possibilistic, and strong — grading the *strength of the obstruction* to a global section of a presheaf of probability distributions. Our filtration is complementary: it grades *what the algebra can support*, not how badly a global section fails. Their axis measures obstruction strength; ours measures coherence demands.

# 4 Discussion

The standard philosophical debate between scientific realism and constructive empiricism treats the question as uniform: either theories aim at truth or they aim at empirical adequacy. The algebraic analysis above lets us recast that choice as *parametrised* by the observation algebra. For Boolean observations, the mathematics does not prefer one position over another. For non-Boolean observations ($\dim \geq 3$), VDR is algebraically impossible; the constructive empiricist position [37] is the generic one, and value-definite realism is the special case — the case in which the observation algebra happens to be distributive.

That value-definiteness is *separable* from realism — that the realist is not, qua realist, entitled to simultaneous definite values, and that what underwrites them is the Boolean (distributive) structure of jointly performable observations — is not new with us. It is the conclusion Stairs [36] reached directly: the requirement that all magnitudes carry classically definite values constrains the admissible structures for reasons independent of realism, turning on Boolean representability rather than on realist commitment; his realism-without-value-definiteness position is, in the present vocabulary, the PR layer with $\mathcal{S}_{\mathrm{df}}$ empty. The parallel link between realism and distributivity is the Putnam [33]–Dummett battleground (Putnam revises logic to keep value-definiteness; we instead read its loss off the algebra), and the primacy of the Boolean/non-Boolean event structure over any hidden dynamics is the Bub–Pitowsky [5] thesis, itself continuous with Pitowsky's reading of quantum mechanics as a non-classical theory of probability on the projection lattice [31]; the broader algebraic-foundations setting is surveyed in Landsman [27]. What the filtration adds to these is not a new philosophical position but an organising one: it locates EA, $\mathrm{PR}_{\mathrm{lat}}$, $\mathrm{PR}_{\mathrm{dual}}$, and VDR on a single axis indexed by distributivity, and separates the obstructions (pasting, regularity, sharpness) that the debate treats as one.

Several classical positions translate directly into this vocabulary, which is offered as a uniform restatement rather than a sharpening of any one of them. The EPR argument for completeness [14] amounts to the demand that all observables simultaneously possess definite values — i.e., VDR — which the Kochen–Specker theorem blocks algebraically for $\dim H \geq 3$. Bohr's complementarity corresponds to the non-triviality of the pasting transition: incompatible contexts cannot be simultaneously realised. Bell's beables [20] — the observables that *can* have definite values — are the elements of the maximal Boolean subalgebra $D(\rho, R)$ (Theorem 2.11), restating Bub–Clifton in this vocabulary. De Finetti's operationalism is EA in the Boolean case; Pitowsky [30] develops the quantum analogue via Dutch book arguments on $L(H)$.

The analysis is not specific to quantum mechanics. Any collection of individually Boolean observation contexts — jointly performable measurements in behavioural science [13], jointly queryable attributes in machine learning [10], simultaneously administrable treatments in experimental design — produces an observation algebra via categorical gluing [18]. If the contexts are mutually compatible, the colimit is Boolean and VDR is available. If not, the colimit is orthomodular and VDR may be blocked. In principle this suggests a diagnostic: when a deterministic latent-state model fails, the failure may be algebraic — the observation algebra does not support VDR — rather than a consequence of noise or insufficient data. We state this only to mark its limits. By the equivalence used above[6] "the failure is algebraic, not statistical" is the same diagnostic as "the failure is contextuality, not nuisance," which the Contextuality-by-Default programme [13, 8] already operationalises on empirical data — and operationalises in the harder regime, since the sheaf/orthomodular formulation requires consistent connectedness (no marginal-selectivity violation), whereas real data routinely violates it and Contextuality-by-Default is built to handle that case. The contribution here is therefore not a new empirical test; it is the placement of that already-finitary diagnostic within the filtration, where the one

[6]Non-distributivity of the colimit is equivalent to the non-existence of a joint distribution, i.e. to contextuality as failure of a global section [1, 16]; the categorical form of this equivalence, orthomodularity as the obstruction to a global section, is [18].

obstruction it cannot see — the $\sigma$-additive, dual-space ($\mathrm{PR}_{\mathrm{dual}}$) layer, which is infinitary — becomes visible as a distinct transition.

Several directions remain open:

1. *The OML extension problem.* Does every state on a general OML extend to a $\sigma$-additive measure on $S_0(A)$, as in the Boolean case? For $L(H)$ the answer is *negative* for *every* state: Gleason supplies the lattice measure but no dual-space charge exists — the Kochen–Specker / Pitowsky non-extendability of §2.1 settles the normal states, and a finite family of infinite-dimensional, pairwise-meet-zero subspaces (a copy of $\mathrm{MO}_2$ with infinite-dimensional atoms) settles the singular states. The general OML case remains open.

2. *Dynamical structure.* If the observation algebra carries a group action (stationarity, exchangeability), how does symmetry interact with the coherence hierarchy? The Boolean case is partially understood (de Finetti's theorem for exchangeability); the OML case is unexplored.

3. *Causal inference.* Holland's fundamental problem of causal inference [22] — that both potential outcomes $Y(0)$ and $Y(1)$ cannot be observed for the same unit — is structurally analogous to Bohr complementarity. Whether partial identification constraints generate non-Boolean algebraic structure, and whether the commensurability theorem applies to counterfactual reasoning, is unexplored.

4. *Relation to the topos programme.* The Bohrification programme [24, 12, 21] encodes related structure more powerfully: the spectral presheaf over the poset of Boolean subalgebras captures VDR-failure as absence of global sections (Kochen–Specker), and local VDR within each Boolean context is built into the presheaf's local sections. Our filtration operates at a coarser level — distinguishing EA, PR, and VDR as properties of the algebra's state space rather than as sheaf-cohomological obstructions. The McDonald–Bimbó duality offers a complementary perspective: a single topological dual space (not a presheaf over contexts) in which principal filters are structurally distinguished. Whether the EA/PR/VDR vocabulary translates into the topos framework, and whether the filtration refines or merely simplifies the sheaf-theoretic picture, remains open.

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